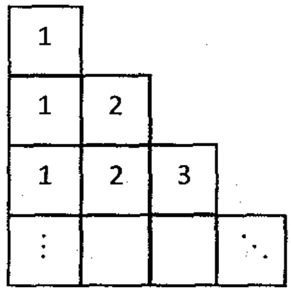
2017 NYJC Prelim Paper 1

1 A board is such that the n^{th} row from the top has n tiles, and each row is labelled from left to right in ascending order such that the i^{th} tile is labelled i, where n and i are positive integers.



Given that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$, by finding the sum of the numbers in the r^{th} row, show that the sum of all the numbers in n rows of tiles is $\frac{1}{6}(n)(n+1)(n+2)$. [4]

2 The curve C has equation $2x - y^2 = (x+y)^2$.

(i) Find the equations of the tangents to C which are parallel to the x-axis. [4]

(ii) The line l is tangent to C at $A(2,-2)$. If the normal to C at the origin O meets l at the point B, find the area of triangle OAB. [4]

3 **Do not use a calculator in answering this question.**

(i) Explain why the equation $z^3 + az^2 + az + 7 = 0$ cannot have more than two non-real roots, where a is a real constant. [1]

(ii) Given that $z = -7$ is a root of the equation in **(i)**, find the other roots, leaving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \le \pi$. [4]

(iii) Hence, solve the equation $iz^3 + 8z^2 - 8iz - 7 = 0$, leaving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \le \pi$. [2]

4 (i) By using the substitution $x-1=3\tan\theta$, find $\displaystyle\int \frac{1}{\sqrt{x^2-2x+10}}\,dx$. [5]

(ii) By expressing $x+3=A(2x-2)+B$, find $\displaystyle\int \frac{x+3}{\sqrt{x^2-2x+10}}\,dx$. [3]

5 (i) By considering $f(r)-f(r+1)$, where $f(r)=\dfrac{\sqrt{r}}{2\sqrt{r}+1}$, find

$$\sum_{r=1}^{n} \frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}$$

in terms of n. [3]

(ii) Hence, find $\displaystyle\sum_{r=1}^{\infty} \frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}$. [2]

(iii) Find the smallest integer n such that

$$\sum_{r=1}^{n} \frac{\sqrt{r+1}-\sqrt{r+2}}{\left(2\sqrt{r+1}+1\right)\left(2\sqrt{r+2}+1\right)} < -0.1.$$ [3]

6 The curve C has equation

$$y=1+\frac{2x+p}{(x-2)(x+3)},$$

where p is a constant.

(i) Find the range of values of p for which C has more than one stationary point. [4]

(ii) Sketch C for p = 7, stating the coordinates of the turning point(s) and the points of intersection with the axes and the equations of any asymptotes. [3]

(iii) By sketching a suitable graph on the same diagram, solve the inequality

$$1+\sqrt{12-x^2} \geq \frac{2x+7}{(2-x)(x+3)}.$$ [3]

7 The functions f and g are defined by

$$\mathrm{f}: x \mapsto \mathrm{e}^{-x^2}, \quad x \in \mathbb{R},\ x<0,$$

$$\mathrm{g}: x \mapsto \frac{1}{x+3}, \quad x \in \mathbb{R},\ x \neq -3.$$

(i) Show that g^{-1} exists, and define g^{-1} in a similar form. [2]

(ii) State the solution set for $\mathrm{g}\mathrm{g}^{-1}(x)=x$. [1]

(iii) Explain why fg^{-1} does not exist. [1]

Let the function h be defined by

$$\mathrm{h}: x \mapsto \mathrm{g}(x), \quad x \in \mathbb{R},\ x<k,$$

where k is a real constant.

(iv) Given that $\mathrm{f}\,\mathrm{h}^{-1}$ exists, state the maximum value of k. [1]

(v) For the value of k found in **(iv)**,

(a) find the exact range of $\mathrm{f}\,\mathrm{h}^{-1}$, [2]

(b) solve $\mathrm{h}(x)=\mathrm{h}^{-1}(x)$. [2]

8 A curve C has parametric equations

$$x=1+\mathrm{e}^{t}+\mathrm{e}^{-t}, \quad 2y=\mathrm{e}^{t}-\mathrm{e}^{-t}, \quad t \in \mathbb{R}.$$

(i) Show that the Cartesian equation of C is $\dfrac{(x-1)^2}{2^2}-y^2=1$. [2]

(ii) Sketch C, showing clearly the equations of any asymptotes and coordinates of the centre and the point(s) where the curve cuts the x-axis. [3]

(iii) Find the exact area of the region bounded by C and the line $x=1+\mathrm{e}+\mathrm{e}^{-1}$. [4]

(iv) Find the volume of the solid of revolution when the region bounded by C and the lines $x=3$ and $y=4$ is rotated completely about the y-axis. [2]

9 With reference to an origin O, a particle P moves in space with position vector $(\lambda-\mu)\mathbf{i}+(1+2\mu)\mathbf{j}+(2-3\lambda)\mathbf{k}$. Another particle Q moves along the line l with equation $\mathbf{r}=\begin{pmatrix}1\\1\\0\end{pmatrix}+t\begin{pmatrix}2\\-2\\-3\end{pmatrix}$, $t\in\mathbb{R}$.

(i) State the locus of P. [1]

(ii) Determine if the particles P and Q can meet. [3]

(iii) Find the shortest possible distance between P and Q. [2]

Another particle R moves along the line m with equation $\mathbf{r}=\begin{pmatrix}1\\k\\0\end{pmatrix}+s\begin{pmatrix}2\\-2\\-3k\end{pmatrix}$, $s\in\mathbb{R}$, where k is a constant.

(iv) Find condition(s) satisfied by k if lines l and m are skew lines. [3]

(v) A particle is shot from $X(0,-1,-5)$ perpendicularly toward the path of Q. Find the coordinates of the point where it crosses the path of Q. [2]

10 A car is travelling at a speed of 30 m/s on a road heading towards a perpendicular train track, which is elevated 30 m above the ground. The front of the car is 40 m away from the track when the front of the train first crossed the road.

If the train is travelling at 20 m/s, show that the distance between the front of the train and the car is $\sqrt{1300t^2-2400t+2500}$ m. [2]

(i) How fast is the front of both the train and the car separating 1 second later? [2]

(ii) Find the distance when the front of the train and the front of the car are closest. [4]

(iii) Find the rate of change of the angle of elevation of the front of the train from the car 1 second later. [4]

11 Suppose a point P on the rim of a wheel of radius r is initially at the point O. As the wheel roll along the x-axis without slippage, the locus of P, known as a *cycloid*, has parametric equations given by

$$x=r(\theta-\sin\theta),\ y=r(1-\cos\theta),\ \theta\geq 0.$$

(i) Sketch the locus of P for $0\leq\theta\leq 4\pi$. [2]

(ii) Show that $\dfrac{dy}{dx}=\cot\dfrac{\theta}{2}$. [3]

(iii) Show that the curve is a solution to the differential equation $\left(\dfrac{dy}{dx}\right)^2=\dfrac{2r}{y}-1$. [3]

(iv) Find the exact area bounded by the locus of P and the x-axis for $0\leq x\leq 2\pi r$. [4]

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